W-B-131M
July 30, 1984
Superseding W-B-131
January 16, 1978

FEDERAL SPECIFICATION
BATTERY, STORAGE (VEHICULAR, IGNITION LIGHTING, AND STARTING)

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers vehicular storage batteries of the lead acid type.

1.2 Classification. Vehicular storage batteries furnished under this specification shall be of the SAE/BCI battery type numbers specified in Table IV and identified by a type designation number (see 1.2.1).

1.2.1 Type designation number. Venicular storage batteries covered by this specification shall be identified by a type designation number formed as indicated below. This type designation number is intended for cataloging and ordering purposes and not for surface printing on the vehicular battery.

W-B-131	-	5	-	71-275	-	F
Spec. No. (1.2.2)		Class of Battery (1.2.3)		SAE Type No. (1.2.4)		Maintenance-free Wet or Dry Charged (1.2.5)

1.2.2 Vehicular battery specification No. W-B-131.

1.2.3 Class of vehicular batteries shall be designated as follows:

1 - Class 1 - Passenger car and light commercial, 12 volt.
2 - Class 2 - Heavy duty commerical, 12 volt.
3 - Class 3 - Passenger car and light commercial, 6 volt.
4 - Class 4 - Heavy duty commercial, 6 volt.
5 - Class 5 - Maintenance-free passenger car and light commercial, 12 volt.
6 - Class 6 - Maintenance-free heavy duty commercial, 12 volt.
7 - Class 7 - Maintenance-free cycling service, 12 volt.

1.2.3 The SAE Type Designation No. shall consist of the types specified in Table IV.

1.2.5 Designation of dry, wet charged, or maintenance free batteries shall be as follows:

D - Dry charged.
W - wet charged.
F - Maintenance-free

2. APPLICABLE DOCUMENTS

2. The following documents of the issues in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Federal Specification:

PPP-B-140 Batteries, Storage, Industrial, Packaging of.

Federal Standard:

Fed Std No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 6140

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(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; SanFrancisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

MILITARY STANDARDS:

MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect of date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

D 639 - Testing Battery Containers Made from Hard Rubber or Equivalent Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1961 Race Street, Philadelphia, PA 19103.)

Battery Council International (BCI):

BCI Battery Technical Manual

(Application for copies should be addressed to Battery Council International, 111 East Wacker Drive, Chicago, IL 60601.)

Society of Automotive Engineers, Inc., Publication:

SAE Handbook - Storage Batteries (J537) SAE standard.
- Life Test for Automotive Storage Batteries - Recommended Practice SAE J240a.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15906.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street, N.W. Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Warranty. The manufacturers' standard commercial warranty or the warranty specified in Federal Supply Schedule Group 61, Part 1, batteries: 6 or 12-volt lead-acid storage shall apply.

3.2 Test report. Each manufacturer or distributor shall furnish certified test reports acceptable to the contracting officer that their batteries meet the requirements of Section 3 for each battery designation. The solicitation shall designate on which battery a certified test report is required. The certified test report shall be dated within two years before the solicitation date.

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3.3 Materials. All material used in the construction of batteries purchased under this specification shall conform to standards of the best commercial practices. Active material, grid alloy, or miscellaneous lead parts smelted from other batteries must be refined to remove impurities prior to subsequent use in new batteries. Materials not specifically designated shall withstand the operational and environmental test specified herein without adversely affecting battery performance.

3.4 Design and construction.

3.4.1 Dimensional limitations. Batteries shall be within the dimensional limitations specified in SAE J537.

3.4.2 Containers. Containers shall be free of leaks, cracks, or other defects that would adversely affect performance. Containers shall be of one piece construction and shall be made of non-absorbent, acid-resistant, hard rubber or polypropylene plastic meeting the physical, chemical, and electrical requirements specified (see 3.6.1).

3.4.3 Cell covers. Covers shall meet the same physical, chemical, and electrical requirements as the container material (see 3.6.1.3). Cell covers may be one-piece or multiple cell construction.

3.4.4 Sealing compund and heat sealing. The seal resulting from sealing by compound or heat shall be acid-resisting and shall maintain an unbroken seal between the cover and the container throughout the temperature range 0 degrees to 150 degrees F., and there shall be no adhesion or cohesion failures under any of the test conditions specified herein.

3.4.5 Filler plugs and electrolyte level.

3.4.5.1 Classes 1, 2, 3, and 4 of vehicular batteries. The cell covers shall provide openings for each cell for filling plugs and the observation of electrolyte level. Normal electrolyte level shall be not less than 3/8 inch above the top of the separators and be easily identifiable. The plugs may be of the screw, bayonet, gang or push-in type and shall incorporate baffles or other means of minimizing the splashing of electrolyte. The material in the plugs shall be non-absorbent, acid-resistant, and have no adverse effect on the electrolyte. By design, the filler plugs shall incorporate an anti-flame propagation device and be vented.

3.4.5.2 Classes 5, 6, and 7 of maintenance free batteries. Maintenance free batteries should not require the addition of water during normal service life, provision for the anti-flame propogation device and gas venting shall be accomplished by design, but openings in the battery cover to allow the routine addition of water need not be provided.

3.4.6 Terminal seals. Terminal seals shall be provided to prevent leakage between terminals and battery cover or container and shall be acid-resistant.

3.4.7 Metal parts. All metal parts, other than lead alloys, shall be coated with lead, or otherwise treated to resist acid.

3.4.8 Internal construction.

3.4.8.1 Plates. Plates used in construction of batteries used in this specification shall be of the size, number, and thickness that will enable the battery to meet its electrical performance requirements as specified in table IV.

3.4.8.2 Intercell connectors and post straps. Intercell connectors shall be designed for the required performance. In addition, the intercell connectors of 12-volt batteries shall not be exposed (excluding batteries recommended for motor coach or diesel engine use)

3.4.8.3 Separators. The separators shall be of material that will enable the battery to meet all the performance requirements specified in Table IV.

3.4.8.4 Double insulation. When specified (see 6.2), double insulation shall be furnished in batteries of the types in which double insulation is commercially available. The double insulation shall be furnished in batteries recommended for motor coach or bus service. When a battery is built with double insulation, the cold activation, reserve capacity, and cold cranking requirements shall be reduced 15 percent from the values listed in Table IV.

NOTE: For the purpose of this specification, double insulation is defined as the use of a retaining sheet of perforated, porous material, including glass mats, between the positive plate and the customary single separator.

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3.4.9 Terminals. Terminals shall be located as specified on the applicable BCI assembly figure number of Table IV. The polarity markings shall be in accordance with SAE J537. The terminals shall be in accordance with SAE J537, Figures 1, 2, 3, and 3A.

3.4.10 Electrolyte. Unless otherwise specified (see 6.2), wet charged batteries shall be furnished with electrolyte that at full charge has a specific gravity of 1.265 ± 0.010 at 80 degrees F at time of shipment. Unless otherwise specified (see 6.2), electrolyte shall not be furnished with batteries shipped dry charged.

3.4.11 Maintenance free. When specified, batteries shall be furnished as maintenance-free wet units having the following characteristics.

3.4.11.1 Internal construction. Electrolyte reserve shall be such that watering of the batteries is not required during normal service life.

3.4.11.2 External construction and venting system. The battery cover and vent system shall be designed to prevent electrolyte loss during service and to keep the top of the battery free from electrolyte.

3.4.11.3 Side terminal batteries. Side terminals, where provided, shall be of the type which are to be mated with sealed terminal connectors to tightly secure and seal the termination from corrosion and thus assure a maintenance-free connection.

3.4.12 Cycling service.

3.4.12.1 Maintenance free. When specified, the maintenance free units shall contain plate thickness and formulation suitable for withstanding the cycling type of service present in emergency, law enforcement, school bus, and delivery - type vehicles as well as other cycling applications. Maintenance free cycling service type batteries shall meet cycle life requirements of 3.7.6.

3.4.12.2 Nonmaintenance free. When specified, the nonmaintenance free units shall be of the High Cycle type and contain plate thickness and formulation suitable for withstanding the cycling type of service present in emergency law enforcement, school bus and delivery type vehicles as well as other cycle installation. Plates and separators shall meet vibration and other stresses. Nonmaintenance free High Cycle type batteries shall meet cycle life requirements of 3.7.6.

3.5 Condition for shipment. Batteries purchased under this specification shall be shipped in one of the following conditions, as specified (see 6.2).

3.5.1 Batteries intended for immediate use of or wet storage with charging equipment. Wet charged batteries shall be filled with electrolyte (see 3.4.11) and shall be fully charged.

3.5.2 Batteries intended for storage in the unfilled state and without charging. The following condition is acceptable where storage in the unfilled state is intended.

3.5.2.1 Dry charged. Plates and separators of dry charged batteries shall meet the performance requirements when activated. The vent hole for each cell shall be closed by easily removeable, air-tight, and moisture-proof type seals or appropriate tape, which shall be held firmly in place until removed. Seals for vent holes are not required when the battery is enclosed in a six-sided box and sealed with weather resistant tape along the seams. Alternately, seals for vents are not required when the battery is enclosed and sealed in an air-tight and moisture-proof bag.

3.5.3 Maintenance free. Maintenance free batteries shall be filled with electrolyte (see 3.4.11) and shall be fully charged.

3.6 Physical.

3.6.1 Containers.

3.6.1.1 Electrical breakdown. When tested as specified in 4.3.2.1.1, the container shall be free of leaks and other imperfections and shall show no evidence of being perforated or burned through.

3.6.1.2 Acid resistance. When tested as specified in 4.3.2.1.2, the material shall not increase in weight more than 1.5 percent or increase dimensionally greater than 2 percent. It shall show no evidence of blisters, cracks, or other visible damage.

3.6.1.3 Impact resistance. When tested as specified in 4.3.2.1.3 and 4.3.2.2, the minimum impact resistance required for containers and covers shall be as in table I.

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3.6.1.3.3 Impact resistance. The minimum impact resistance values as indicated in table I shall apply to a test temperature of 80 degrees ± 5 degrees F. This test temperature is applicable to both hard rubber and plastic materials.

TABLE I

Battery Service (From Table I)	Minimum Impact Resistance	
	Hard Rubber	Plastic
Classes 1,3,5,6, and 7	12 In. lb Avg. for short spans 12 In. lb. Avg. for long spans No impact failure below 8 In. lb.	50 In. lb. Avg. for short spans 50 In. lb. Avg. for long spans No single impact failure below 35 In. lb.
Classes 2 and 4	20 In. lb. Avg. shorts spans 20 In. lb. Avg. long spans No impact failure below 16 In. lb.	75 In. lb. Avg. for short spans 75 In. lb. Avg. for long spans No single impact failure below 50 In. lbs.

3.6.1.4 Bulge test. When tested as specified in 4.3.2.1.4, the bulge resistance of a hard rubber container shall not exceed 1/16 inch and for a plastic container shall not exceed 1/8 inch.

3.6.2 Covers. Cover material shall meet the minimum impact requirements specified for battery containers (see 3.6.1.3 and 4.3.2.2).

3.6.3 Leakage. When tested as specified in 4.3.2.3. each cell shall independently maintain a pressure of two pounds per square inch (psi) for a period of five seconds with a maximum loss of 0.1 psi. Evidence of any physical defect in the container, cover or seal between cover and container at the specified pressure or inability to attain that pressure shall constitute a failure. All failed batteries shall be rejected.

3.6.4 Vibration. When tested as specified in 4.3.2.4, 4.3.2.4.1, and 4.3.2.4.2, each battery shall survive 2 units of vibration, there shall be no flooding of electrolyte, the battery shall meet the leakage requirements of 3.6.3, the battery shall meet the cold cranking requirement for 0 degrees F., and on external and internal examination there shall be no evidence of mechanical defects or leaks.

3.7 Electrical performance.

3.7.1 Cold activation at 30 degrees F. (dry charged batteries only). When tested as specified in 4.3.3.1, the battery terminal voltage under load at the end of 15 seconds discharge shall be equivalent to 1.2 volts per cell or greater.

3.7.2. Reserve capacity and cold cranking. The reserve capacity and cold cranking shall be in accordance with the BCI test sequence.

3.7.2.1 Reserve capacity. When tested in accordance with 4.3.3.4, 90% of the batteries tested shall meet or exceed the specified value in Table I,and 100% of the batteries shall meet or exceed 90% of the specified value in Table I.

3.7.2.2 Cold cranking performance at 0 degrees F. When tested in accordance with 4.3.3.5, the battery shall meet the performance requirements specified in SAE Standard J537.

3.7.3 Charge rate acceptance test. When tested in accordance with 4.3.3.3, the minimum acceptance is 2 percent of the cold cranking rating shown in Table IV.

3.7.4 Retention of charge.

3.7.4.1 Classes 1, 2, 3, and 4 of vehicular batteries. When tested as specified in 4.3.3.6, the battery shall yield as a minimum, 75 percent of the reserve capacity specified in Table IV.

3.7.4.2 Classes 5, 6, and 7 of maintenance free vehicular batteries. When tested as specified in 4.3.3.6, the battery shall yield as a minimum, 90 percent of the reserve capacity specified in table IV.

3.7.5 Dry cell internal resistance (dry charged batteries only). When tested as specified in 4.3.3.7, the average terminal-to-terminal resistance of each cell in both directions, after assembly in the container, shall be no less than 50,000 ohms.

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3.7.6 Cycle life. When tested as specified in 4.3.3.9, the class 7 battery shall yield a minimum total cycles equal to 50 times its reserve capacity minute rating.

3.8 Identification marking. The following data shall be molded on the sides, the top of the cover, or applied to a permanent electrolyte resistant label or plate securely and permanently attached to the battery container and contain the following information.

Battery Identification Number
Cold Cranking at 0 degrees F. Rating (the test name or abbreviation thereof followed by the rating)
Manufacturer's Name or Identification

Date of manufacture (month, year) and plant of manufacture, if the same battery is manufactured at more than one plant, shall be legibly stamped or branded on the outside cell connector, top or side of the cover or container. A plate or label permanently affixed to the top or side of the outer cover or container, or a tag permanently affixed to the terminal post is acceptable. Coded information is acceptable as long as the key to the code is made available.

3.9 Instruction for placing battery into service. Instructions for placing the battery into service, as required, shall be provided with each battery. The instructions shall show the latest recommended date for placing the battery in service (if required) and shall include the specific gravity of electrolyte for filling (charged and dry) and charging instructions. The method of providing the instructions shall be in accordance with the manufacturer's standard practice.

3.10 Workmanship. Containers, covers, and vent caps shall be free from cracks, lead, and broken parts. Lead burning or welding shall be homogeneous and free from blow holes or imperfect bonds between parts which have been burned or welded together. Handles and metallic parts shall be free of burrs, sharp edges, and corrosion. Marking shall be clear and distinct. The sealing compound shall not be cracked or separated from the holding surfaces.

3.11 Dry charged. After activation in accordance with manufacturer's instructions, batteries shall meet the performance requirements specified in able IV.

4. Quality Assurance Provisions.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Acceptance inspection.

4.2.1 Sampling for acceptance inspection. Unless otherwise specified, sampling shall be performed in accordance with the provisions set forth in MIL-STD-105 inspection level S-2.

4.2.2 Inspection lot. Inspection lot shall consist of batteries of one identification number from an identifiable production period, from one manufacturer, and one plant, submitted for acceptance.

4.2.3 Dimensional and visual examination and acceptable quality level. Each battery selected in accordance with 4.2.1 shall be examined for defects in accordance with table IV. The acceptable quality levels (AQL) on the basis of percent defective shall be as follows in table II. Classification should be on a total basis for the 4.0 AQL.

Table II

Defect Classification	AQL
Major	1.5
Minor	4.0

4.2.3.1 Classification of defects. For examination purposes, defects shall be classified as specified in table III.

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Table III Classification of Defects

Categories	Defects	Method of Inspection
Major	Maximum dimensional limitation exceeded (see 3.4.1).	Scale
	Leaks or cracks in container	Visual
	Covers not properly sealed to containers (see 3.4.4).	Visual
	Filler plugs and openings not as specified (see 3.4.5).	Visual
	Terminal seals not as specified (see 3.4.6).	Visual
	Location and polarity of terminal post not as (see 3.4.9).	Visual
	Terminal markings not as specified (see 3.4.9).	Visual
	Dimension of terminal post not as specified (see 3.4.9).	Gauge
	Specific gravity of electrolyte out of limits charged (Hydrometer and wet batteries only) (see 3.4.10 and 3.5.1).	
	Low electrolyte level (charged and wet only) (see 3.5.1).	
	Vent holes not properly sealed (as applicable) (see 3.5.2).	Visual
	Loose terminal posts.	Visual
Minor	Improper marking (see 3.8).	Visual
Minor	Missing instructions (see 3.9).	Visual
Minor	Burrs of sharp edges (see 3.10).	Visual

4.2.4 Test for acceptance. Each month during the contract while the batteries are in the process of manufacture, batteries from each plant shall be tested for cold activation, reserve capacity, and cold cranking at 0 degrees F. (see 4.3.3.1, 4.3.3.4, and 4.3.3.5). The sample for testing shall be in accordance with MIL-STD-105 inspection level S-1 from each class. The applicable AQL shall be 6.5. percent defective for the tests in 4.4.3.1 and 4.4.3.5. Any battery failing to pass either 4.3.3.1 or 4.3.3.5 shall be considered defective. If the sample does not meet the average reserve capacity requirement it shall be considered to have failed. In case of failure, the contractor shall suspend offering the battery types until corrective measures, as necessary, have been taken and another sampling, using the single sampling plan for tightened inspection, has been tested and passed.

4.3 Test procedures.

4.3.1 General.

4.2.1.1. Conditions. Tests shall be conducted under general laboratory conditions. Unless otherwise specified herein, tests shall be conducted at ambient room temperature of 80 degrees ± 5 degrees F.

4.3.2 Physical.

4.3.2.1 Containers.

4.3.2.1.1 Electrical breakdown. The electrical breakdown test shall be conducted as specified in ASTM D639 to determine conformance to 3.6.1.1. The voltage shall be 100 v/mil for hard rubber and 300 v/mil for plastic.

4.3.2.1.2 Acid resistance. The acid resistance test shall be conducted as specified in ASTM Standard D639 to determine conformance to 3.6.1.2.

4.3.2.1.3 Impact resistance. The impact resistance test for hard rubber containers shall be conducted as specified in ASTM D639 to determine conformance to 3.6.1.3 with impact points and number of impacts as specified in the BCI Battery Technical Manual. The impact resistance test for plastic containers shall be conducted as specified in Plastic COntainer COver, Testing Procedures BCI Battery Technical Manual, to determine conformance to 3.6.1.3. The test temperature of plastic container shall be 80 degrees ± 5 degrees F.

4.3.2.1.4 Bulge resistance (container only). The bulge resistance test for hard rubber containers shall be conducted as specified in ASTM D639 to determine conformace to 3.6.1.4. The bulge resistance test for plastic containers shall be conducted as specified in Plastic Container and Cover Testing Procedures, BCI Battery Technical Manual, to determine conformance with 3.6.1.4. The test temperature for plastic covers shall be 80 degrees ± 5 degrees F.

4.3.2.2 Impact resistance (covers).

4.3.2.2.1 Hard rubber covers. Three flat plate cell cover material specimens shall be molded and tested according to ASTM D639.

4.3.2.2.2. Plastic one-piece covers. Three whole covers shall serve as test specimens. The impact resistance test for covers shall be conducted as specified in Plastic Container and Cover Testing Procedures, BCI Battery Technical Manual, to determine conformance to 3.6.1.3. The test temperature for plastic covers shall be 80 degrees ± 5 degrees F.

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4.3.2.3 Leakage test. Two pounds per square inch shall be applied to the battery to establish compliance to 3.6.3.

4.3.2.4 Vibration. To determine conformance to 3.6.4.

4.3.2.4.1 Batteries with hard rubber cases. The batteries shall be tested as specified in SAE Standard J537, "Vibration test".

4.3.2.4.2 Batteries with plastic cases. The vibration test for plastic case batteries shall be as specified in SAE Standard J537, "Vibration test."

4.3.3 Electrical. The electrical tests shall be conducted in the sequence as listed below.

4.3.3.1 Cold activation test. This test shall apply to dry charged. The test shall be conducted as specified in the BCI Battery Technical Manual.

4.3.3.2 Conditioning. Perform cold activation test, then condition the battery and charge it per SAE J537 before conducting the remaining electrical tests.

4.3.3.3 Charge rate acceptance test. To determine conformance to 3.7.4, the charge rate acceptance test shall be conducted as specified in SAE Standard J537.

4.3.3.4 Reserve capacity rating. To determine conformance to 3.7.2.1, the reserve capacity test shall be conducted as specified in SAE Standard J537.

4.3.3.5 Cold cranking test at 0 degrees F. To determine conformance to 3.7.2.2, the cold cranking test of 0 defrees F shall be conducted as specified in SAE Standard J537.

4.3.3.6 Retention of charge. To determine conformance to 3.7.4, the battery shall be conditioned in accordance with 4.3.3.2 and allowed to stand on open circuit for 4 weeks in an ambient temperature of 75 degrees F $\pm$ 5 degrees F. At the end of this period and without recharging, the battery shall be discharged at the reserve capacity rate in accordance with 4.3.3.4.

4.3.3.7 Dry cell internal resistance. To determine conformance to 3.7.5, the resistance shall be measured with an ohm-meter, bridge, or other test instrument.

4.3.3.8 Inspection of preparation for delivery. Inspection of preparation for delivery shall be accomplished in accordance with PPP-B-140.

4.3.3.9 Cycle life. To determine conformance with 3.7.6, the battery shall be cycled as specified in SAE Standard J240.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The packaging shall be Level A or commercial as specified in the contract or order.

5.1.1 Level A. Charged, dry and wet batteries shall be preserved and packed in accordance with Level A specified in PPP-B-140.

5.1.2 Commercial. Maintenance-free charged and dry or wet batteries shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment and storage.

5.2 Packing. The packing shall be Level A or commercial as specified in the contract or order.

5.2.1 Level A. Maintenance-free charged and dry batteries shall be packed in accordance with Level A specified in PPP-B-140.

5.2.2 Commercial. Charged, dry and wet batteries packed as specified in 5.1.2, shall be packed in fiberboard boxes to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Unitization/palletization. When specified in the contract or order, the shipping containers shall be unitized/palletized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2500 pounds in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking. Marking shall be in accordance with 5.4.1 or 5.4.2 as specified (see 6.2).

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5.4.1 Civil agencies. Each shipping container and unitized loads (when applicable) shall be marked in accordance with Federal Standard No. 123.

5.4.2 Military activities. Each shipping container and unitized loads (when applicable) shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The batteries covered by this specification are intended primarily for starting, lighting, and ignition service on passenger cars, motor-trucks, and motor-coaches equipped with regulated charging systems, but may be used for other purposes for which they are suitable.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type designation No. (see 1.2.1).
(c) Double insulation, if required (see 3.4.8.4).
(d) Specific gravity of electrolyte, if other than specified (see 3.4.10).
(e) Condition of battery at time of shipment (see 3.4.10, 3.5, and 6.3).
(f) When required, quantity and method of packaging of electrolyte for charged and dry (see 3.4.10).
(g) Level of preservation and packaging and the level of packing required (see 5.1).
(h) Any special marking requirements (see 5.4).
(i) Unitization if required (see 5.3).

6.3 Batteries for immediate use. When batteries are being procured for immediate use, and when so specified (see 6.2), batteries may be procured in the charged and wet conditions.

MILITARY INTERESTS:

Military Coordinating Activity

Navy - AS

Custodians

Army - AT
Navy - YD
Air Force - 99

Review Activity

Army - EL

CIVIL AGENCY COORDINATING ACTIVITIES:

DC GOV'T - DCG
USPS - POS
USDA - AFS

PREPARING ACTIVITY:

GSA-FSS

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TABLE IV Battery classifications and ratings (e).

SAE Battery No.	REF. Battery No.	BCI group size	BCI Assy. fig. No.	Performance rating		Amp-hour at 20-hr rate (a)
				Reserve capacity minutes	Cold crank amp 0°F	
Class 1 - Passenger car and light commercial, 12 volt						
3EE-300	13M2A	3EE	9	90	300	70
22F-350	25MJ1A	22F	11L	88	350	53
22NF-245	18M1	22NF	11F	52	245	40
24-305	9M3B	24	10	75	305	50
24-385	9M3D	24	10	95	385	60
24-430	9M3E	24	10	105	430	61
24-285	9MJ3C	24C	10	75	285	53
24C-410	9MJ3G	24C	10	110	410	62
24F-330	17M2	24F	11F	99	330	55
24F-440	25MJ2	24F	11L	120	440	68
24T-370	9T3	24T	10	105	370	70
27-355	9MC6	27	10	105	355	70
27F-355	17M3B	27F	11F	125	355	70
27F-455	17M3A	27F	11F	136	455	80
27H-435	9H5	27H	10	125	435	70
27R-430	10M7	27R	11	125	430	70
29HR-520	10H10	29HR	11	163	520	85
29NF-330	18M3A	29NF	11F	100	330	65
53-210	14M2A	53	14	40	210	40
60-360	15M4A	60	12	110	360	70
72-275	22MJ1	72	17	60	275	45
74-285	22MJ2	74	17	75	285	50
74-335	22MJ2A	74	17	98	335	60
74-410	22MJ2B	74	17	110	410	65
74-455	22MJ2C	74	17	140	455	73
U1-235	---	U1	10	40	235	--
U1-255	---	U1	10	35	255	--
Class 2 - Heavy duty commercial, 12volt						
30H-460	9H9(b)	30H	10	158	460	90
4D-640	20T4A	4D	8	285	640	150
4D-925	---	4D	8	252	925	---
8D-900	20T8A	8D	8	390	900	200
8D-1000	---	8D	8	430	1000	---
Class 3 - Passenger car and light commercial, 6 volt						
1-475	1M1	1	2	159	475	100
2-560	1H2	2	2	220	560	120
2-725	--	2	2	200	725	--
2E-475	8H2	2E	5	160	475	100
2E-485	8H2	2E	5	160	485	100
3-1310	--	3	5	500	1310	--

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SAE Battery No.	REF. Battery No.	BCI group size	BCI ASSY. fig. No.	Performance rating		Amp-hour at 20-hr rate (a)
				Reserve capacity minutes	Cold crank Amp 0°F	
Class 3 - (continued)						
4-700	1H4	4	2	275	700	150
4-720	---	4	2	300	720	---
4-900	---	4	2	340	900	---
5D-830	---	5D	2	340	830	---
Class 4 - Heavy duty commercial, 6 volt						
7D-900	6T3A	7D	2B	430	900	200
7D-940	---	7D	2	400	940	---
7D-915	---	7D	2	300	915	---
Class 5 - Maintenance free, passenger car and light commercial, 12 volt						
22F-375	---	22F	11F	82	375	50
22F-300	---	22F	11F	50	300	--
22F-330	---	22F	11F	70	330	--
22F-490	---	22F	11F	90	490	--
24-270	---	24	10	67	270	70
24-390	---	24	10J	90	390	--
24-410	---	24	10J	80	410	--
24-470	---	24	10J	125	470	--
24-470/1	---	24	10J	120	470	--
24-500	---	24	10J	125	500	--
24-500/1	---	24	10J	120	500	--
24-500/2	---	24	10J	90	500	--
24-520	---	24	10J	130	520	--
24-520/1	---	24	10J	125	520	--
24-545	---	24	10J	135	545	--
24-545/1	---	24	10J	140	545	--
24-550	---	24	10J	110	550	--
24C-425	---	24C	10J	100	425	--
24C-440	---	24C	10J	100	440	--
24C-500	---	24C	10J	90	500	--
24F-375	---	24F	11F	60	375	--
24F-470	---	24F	11F	120	470	70
24F-475	---	24F	11F	115	475	--
24R-500	---	24R	11J	120	500	--
25-430	---	25	10J	100	430	--
27-550	---	27	10	140	550	80
27F-550	---	27F	11F	120	550	80

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SAE Battery No.	REF. Battery No.	BCI group size	BCI Assy. fig. No.	Performance rating		Amp-hour at 20-hr rate (a)
				Reserve capacity minutes	Cold crank amp 0°F	
			Class 5 (continued)			
25-370	---	25	10J	86	370	—
26-335	---	26	10J	62	335	—
26-405	---	26	10J	75	405	—
33-1050	---	33	11F	165	1050	--
54-310	---	54	19K	60	310	—
55-380	---	55	19K	75	380	--
56-450	---	56	19K	90	450	—
57-310	---	57	21	60	310	--
58-380	---	58	21	75	380	--
58-410	---	58	21	82	410	—
58-450	---	58	21	90	450	--
61-310	---	61	20K	60	310	--
63-380	---	62	20K	75	380	--
63-450	---	63	20K	90	450	--
64-475	---	64	20K	120	475	--
64-535	---	64	20K	105	535	--
65-850	---	65	10	140	850	—
70-310	---	70	17J	60	310	--
70-315	---	70	17J	75	315	—
70-355	---	70	17J	70	355	—
70-405	---	70	17J	75	405	--
71-275	---	71	17	60	275	--
71-350	---	71	17	80	350	--
71-390	---	71	17J	90	390	--
72H-430	---	72H	17	100	430	--
73-465	---	73	17J	115	465	—
74-465	---	74	17	125	465	--
74-500	---	74	17	125	500	--
75-500	---	75	17J	90	500	—
75-630	---	75	17J	90	630	--
76-750	---	76	17J	175	750	--
76-1075	---	76	17J	155	1075	--
77-550	---	77	17	140	550	--
78-515	---	78	17J	115	515	—
78-550	---	78	17J	115	550	--
			Class 6 - Maintenance free batteries, heavy duty commercial, 12 volt			
24-475	---	24	10	120	475	70
27-620	---	27	10	160	620	90
31-475	---	31	18	130	475	80
31-475(c)	---	31	18	130	475	80
31-625	---	31	18	160	625	--

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SAE Battery No.	REF. Battery No.	BCI group size	BCI Assy. fig. No.	Performance rating		Amp-hour at 20-hr rate (a)
				Reserve capacity minutes	Cold crank Amp 0°F	
Class 7 - Maintenance free high cycle batteries, 12 volt						
24-375	---	24	10	115	375	70
74-375	---	74	17	115	375	70
27-480	---	27	10	160	480	90
31-580(c)	---	31	18	180	580	95
31-580	---	31	18	180	580	95
31-580(c)	---	31	18	180	580	105(d)

(a) For reference purposes only.
(b) For impact test, these batteries shall be tested to the requirements of class 1.
(c) Furnished with top threaded terminals.
(d) Ampere hour at 100 hour rate.
(e) Dimensions are in SAE Standard J537.

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APPENDIX I

CROSS REFERENCE OF VEHICLE BATTERIES

NSN	W-B-131M	W-B-131J & L
6140-00-184-3415	W-B-131-4-7D-900-D	W-B-131-4-6T3A-D
190-9828	W-B-131-2-8D-900-D	W-B-131-2-20T8A-D
190-9831	W-B-131-2-4D-640-D	W-B-131-2-20T4A-D
190-9840	W-B-131-3-2E-475-D	W-B-131-3-8H2-D
190-9842	W-B-131-3-2E-485-D	W-B-131-1H2-D
190-8502	---	W-B-131-3-1M3-W
191-8506	W-B-131-3-1-475-D	W-B-131-3-1M1-D
191-8519	W-B-131-3-4-700-D	W-B-131-3-1H4-D
195-5841	W-B-131-3-1-475-W	W-B-131-3-1M1-W
199-4988	W-B-131-4-7D-900-W	W-B-131-4-6T3A-W
199-4991	W-B-131-2-8D-900-W	W-B-131-2-8D-900-W
397-2647	W-B-131-1-24C-410-W	W-B-131-1-9MJ3G-W
397-2658	---	W-B-131-1-15M4-D
397-2662	---	W-B-131-1-17M1A-D
397-2665	---	W-B-131-1-17MJ1C-D
397-2670	---	W-B-131-1-17M2A-D
397-2678	W-B-131-1-74-335-D	W-B-131-1-22MJ2A-D
407-3440	---	W-B-131-1-10H7-D
434-5873	---	W-B-131-3-8H2A-D
434-5878	---	W-B-131-3-1M3-D
434-5883	---	W-B-131-3-1M2-D
434-5884	---	W-B-131-3-1M1A-W
434-5885	---	W-B-131-3-1M1A-D
434-5887	---	W-B-131-3-1H4A-D
434-5893	---	W-B-131-3 1H2A-D
438-0208	---	W-B-131-2-20T4B-W
438-0215	---	W-B-131-2-20T4B-D
438-0299	---	W-B-131-2-9H9A-D
446-6758	W-B-131-1-74-455-D	W-B-131-1-22MJ2C-D
446-6773	W-B-131-1-74-410-D	W-B-131-1-22MJ2B-D
446-6813	---	W-B-131-1-9H3B-D
446-6829	---	W-B-131-1-9M3-D
446-6875	---	W-B-131-1-9M3A-D
446-6891	---	W-B-131-1-9MC3A-D
473-1565	W-B-131-1-24C-410-D	W-B-131-1-9MJ3G-D
504-0968	---	W-B-131-1-11M4A-D
538-9984	W-B-131-1-27-355-D	W-B-131-1-9MC6-D
538-9986	W-B-131-1-24-385-D	W-B-131-1-9M3D-D
539-8488	W-B-131-1-27-355-W	W-B-131-1-9MC6-W
539-8491	W-B-131-3-4-700-D	W-B-131-3-1H4-D
539-8507	---	W-B-131-1-11M6-D
643-0495	W-B-131-1-24-305-D	W-B-131-1-9M3B-D
643-0496	---	W-B-131-3-4M2-W
643-0502	---	W-B-131-1-15M4-D
729-8974	W-B-131-1-27R-430-D	W-B-131-1-10M7-D
822-0397	W-B-131-1-22NF-245-D	W-B-131-1-18M1-D
822-0410	W-B-131-1-53-210-D	W-B-131-1-14M2A-D
822-0414	W-B-131-1-29NF-330-D	W-B-131-1-18M3A-D
822-0415	W-B-131-1-29NF-330-W	W-B-131-1-18M3A-W
822-0419	W-B-131-1-27H-435-D	W-B-131-1-9H5-D

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NSN	W-B-131M	W-B-131J & L
6140-00-836-1279	W-B-131-1-24T-370-D	W-B-131-1-9T3-D
836-1280	W-B-131-1-60-360-D	W-B-131-1-15M4A-D
836-1282	W-B-131-1-3EE-300-W	W-B-131-1-13M2A-W
836-1283	W-B-131-1-3EE-300-D	W-B-131-1-13M2A-D
864-7458	---	W-B-131-3-1H2-D
889-1134	W-B-131-1-24F-330-W	W-B-131-1-17M2-W
984-0068	W-B-131-1-24F-330-D	W-B-131-1-17M2-D
984-0143	W-B-131-2-30H-460-D	W-B-131-2-9H9-D
984-0146	W-B-131-1-27F-455-W	W-B-131-1-17M3A-W
984-0147	W-B-131-1-27F-455-D	W-B-131-1-17M3A-D
984-0149	---	W-B-131-1-13TC2-D
984-0155	W-B-131-1-27F-355-D	W-B-131-1-17M3B-D
984-0158	W-B-131-1-24-430-W	W-B-131-1-9M3E-W
984-0159	W-B-131-1-24-430-D	W-B-131-1-9M3E-D
984-0166	---	W-B-131-1-18M3-W
01-010-3228	W-B-131-1-22F-350-D	W-B-131-1-25MJ1A-D
013-2782	W-B-131-1-22F-350-W	W-B-131-1-25MJ1A-W
013-2783	W-B-131-1-24F-440-D	W-B-131-1-25MJ2-D
014-5979	W-B-131-1-24F-440-W	W-B-131-1-25MJ2-W
031-6873	W-B-131-5-71-275-F	W-B-131-1-71-275
031-6874	W-B-131-5-72H-430-F	W-B-131-1-72H-430
031-6875	W-B-131-5-22F-375-F	W-B-131-1-22F-375
031-6876	W-B-131-5-24F-470-F	W-B-131-1-24F-470
031-6877	W-B-131-5-27-550-F	W-B-131-1-27-550
031-6878	W-B-131-5-27F-550-F	W-B-131-1-27F-550
031-6879	W-B-131-6-24-475-F	W-B-131-1-24-475
031-6881	W-B-131-6-31-475-F	W-B-131-1-31-475
031-6882	W-B-131-6-31-620-F	W-B-131-1-31-620
032-1326	W-B-131-6-31-475-F	W-B-131-1-31-475
032-3618	W-B-131-5-74-465-F	W-B-131-1-31-465
032-3619	W-B-131-5-24-470-F	W-B-131-1-24-470
037-9173	W-B-131-5-74-500-F	W-B-131-1-74-500
111-1016	---	W-B-131-1-22MJ2A-M
111-1017	---	W-B-131-1-22MJ2B-M
112-1056	---	W-B-131-1-22MJ1-M
112-1057	---	W-B-131-1-9M3E-M
112-1058	---	W-B-131-1-22MJ2C-M
112-1059	---	W-B-131-1-17M3A-M
112-1060	---	W-B-131-1-17M2-M
112-1061	---	W-B-131-1-17M1-M
112-1062	---	W-B-131-1-14M2A-M
112-1064	---	W-B-131-1-18M3A-M
112-1065	---	W-B-131-1-20T4A-M
112-1066	---	W-B-131-1-9H5-M
112-1067	---	W-B-131-1-10M7-M
112-1068	---	W-B-131-1-17M3B-M
112-1069	---	W-B-131-1-15M4A-M
112-1070	---	W-B-131-1-13M2A-M
112-1071	---	W-B-131-1-9T3-M
112-1072	---	W-B-131-1-9M3D-M
112-1073	---	W-B-131-1-9M3B-M
112-1074	---	W-B-131-1-6T3A-M

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NSN	W-B-131M	W-B-131J & L
6140-01-112-1075	---	W-B-131-1-1H4-M
112-1076	---	W-B-131-1-1H2-M
112-1077	---	W-B-131-1-1M1-M
112-8336	---	W-B-131-1-25MJ2-M
112-6562	---	W-B-131-1-9MJ3G-M
113-6563	---	W-B-131-1-18M1-M
114-2462	---	W-B-131-1-22MJ2-M
114-2463	---	W-B-131-1-22MJ3B-M
114-2464	---	W-B-131-1-25MJ1A-M
115-5277	---	W-B-131-2-28T8A-M
115-5278	---	W-B-131-1-9M6-M
115-9999	---	W-B-131-1-9H9-M
	W-B-131-5-24-500-F	---
	W-B-131-5-24-500/1-F	---
	W-B-131-5-24-500/2-F	----
	W-B-131-5-25-370-F	---
	W-B-131-5-26-335-F	---
	W-B-131-5-26-405-F	---
	W-B-131-5-33-1050-F	---
	W-B-131-5-65-850-F	---
	W-B-131-5-70-310-F	---
	W-B-131-5-70-315-F	---
	W-B-131-5-70-335-F	---
	W-B-131-5-70-405-F	---
	W-B-131-5-71-390-F	---
	W-B-131-5-73-465-F	---
	W-B-131-5-75-500-F	---
	W-B-131-5-75-630-F	---
	W-B-131-5-76-750-F	---
	W-B-131-5-76-1050-F	---
	W-B-131-5-78-515-F	---
	W-B-131-7-78-550-F	---
	W-B-131-7-24-375-F	---
	W-B-131-7-24-375-F	---
	W-B-131-7-27-480-F	---
	W-B-131-7-31-580-F	---
	W-B-131-7-31-580-F	---
	W-B-131-5-54-310-F	---
	W-B-131-5-55-380-F	---
	W-B-131-5-56-450-F	---
	W-B-131-5-57-310-F	---
	W-B-131-5-58-380-F	---
	W-B-131-5-58-410-F	---
	W-B-131-5-58-450-F	---
	W-B-131-5-61-310-F	---
	W-B-131-5-62-380-F	---
	W-B-131-5-63-450-F	---
	W-B-131-5-64-475-F	---
	W-B-131-5-64-535-F	---
	W-B-131-5-22F-490-F	---
	W-B-131-5-22F-330-F	---

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NSN	W-B-131M	W-B-131J & L
	W-B-131-5-22F-300-F	---
	W-B-131-5-24-550-F	---
	W-B-131-5-24-545-F	---
	W-B-131-5-24-545/1-F	---
	W-B-131-5-24-520-F	---
	W-B-131-5-24-520/1-F	---
	W-B-131-5-24-470-F	---
	W-B-131-5-24-470/1-F	---
	W-B-131-5-24-410-F	---
	W-B-131-5-24-390-F	---
	W-B-131-5-24C-500-F	---
	W-B-131-5-24C-440-F	---
	W-B-131-5-24C-425-F	---
	W-B-131-5-24F-475-F	---
	W-B-131-5-24F-375-F	---
	W-B-131-5-24R-500-F	---
	W-B-131-5-25-430-F	---
	W-B-131-5-78-550-F	---
	W-B-131-2-U1-235-W	---
	W-B-131-2-U1-235-D	---
	W-B-131-2-U1-255-W	---
	W-B-131-2-U1-255-D	---
	W-B-131-2-4D-925-W	---
	W-B-131-2-4D-925-D	---
	W-B-131-2-8D-1000-W	---
	W-B-131-2-8D-1000-D	---

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